

Predicting Titanic Passenger Survival

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Introduction

The sinking of the Titanic was a global shock and tragedy that would go on to become one of the most infamous cruise sinking to ever occur. Marketed as an “unsinkable” ship, over a thousand lives were taken in the tragedy. In this analysis, we will be examining what factors predicted the survival rate of a passenger on the ship. We will be examining the following predictors: Class, Gender, number of Siblings and Spouses on board the ship, numbers of parents and children on board the ship, Fare (converted to modern British Pounds), and Port of Embarkation. The Titanic started its journey in Southampton, England and was set to sail to New York City.

Exploratory Data Analysis

BACKGROUND AND VARIABLES

Our data set consists of 622 randomly sampled passengers provided by Frank Harrell, Department of Biostatistics, Vanderbilt University, <https://hbiostat.org/data/repo/titanic.html>. This data includes the following explanatory variables: PClass: ticket class (1 = 1st, 2 = 2nd, 3 = 3rd) Gender: male or female SibSp: number of siblings + spouses of the individual who are aboard the Titanic Parch: number of parents + children of the individual who are aboard the Titanic Fare: Passenger fare (adjusted to equivalent of modern British pounds) Embarked: Port of Embarkation (C=Cherbourg, Q=Queenstown, S=Southampton)

The response variable in this dataset is: Survived: survived (1) or dead (0)

SUMMARY OF THE RESPONSE LABELS IN THE TRAINING DATASET

```
table(titanic_train$Survived)
```

```
##  
##    0    1  
## 388 234
```

```
prop.table(table(titanic_train$Survived))
```

```
##
```

```
##           0           1
## 0.6237942 0.3762058
```

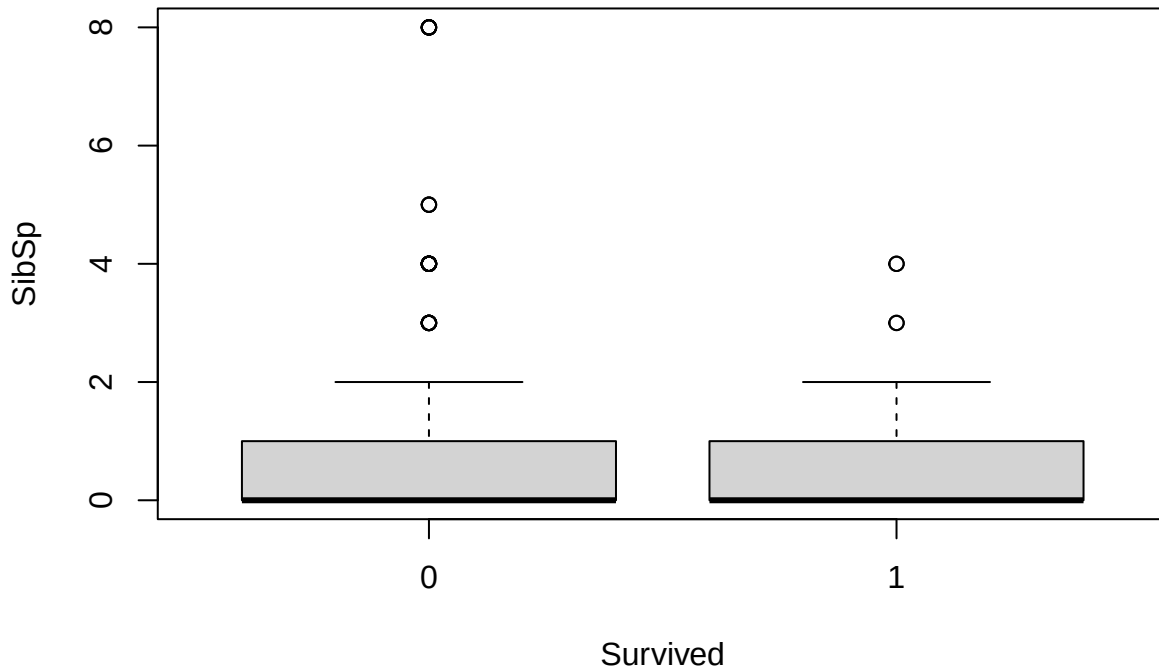
We can see that of all 622 randomly sampled passengers, there was a 37.6% (234 passengers) survival rate, meaning that 62.3% (388) of the sampled passengers did not survive the sinking.

SOME EDA ON RELATIONSHIPS BETWEEN SURVIVAL AND THE QUANTATIVE VARIABLES

In order to help us visualize our explanatory variables and their impact on survival further, we show boxplots below. These will help us in determining if a quantitative predictor variable will be useful in determining the survival of a passenger.

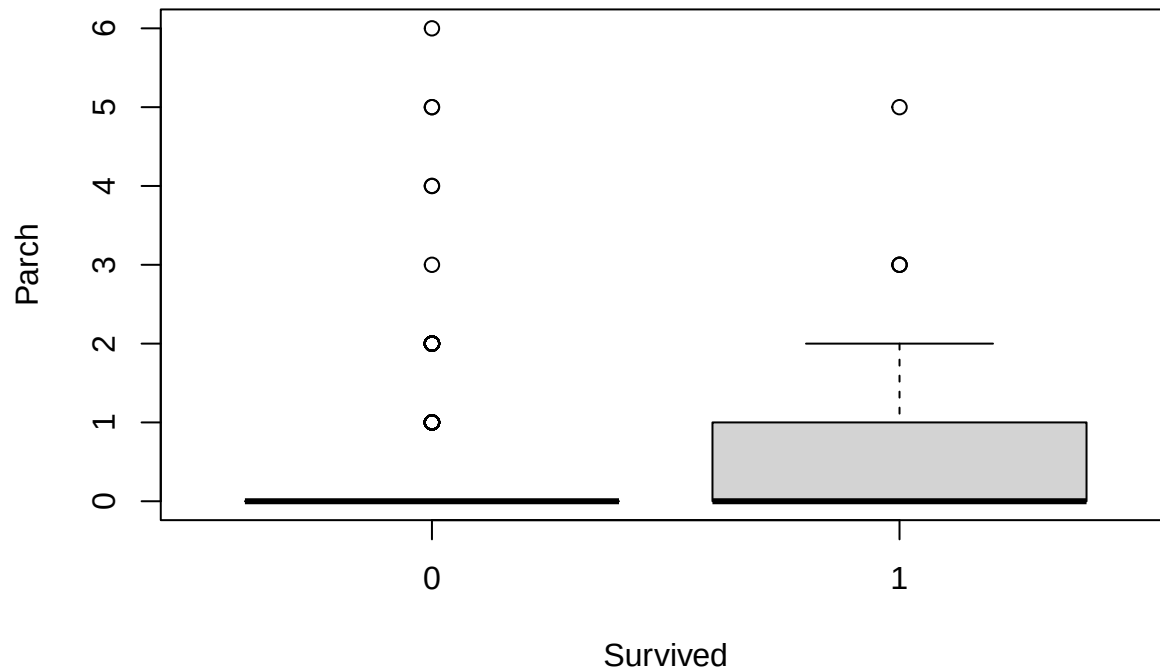
```
boxplot(SibSp ~ Survived,
        data = titanic_train,
        main = 'Total Number of Siblings and Spouses on Board')
```

Total Number of Siblings and Spouses on Board



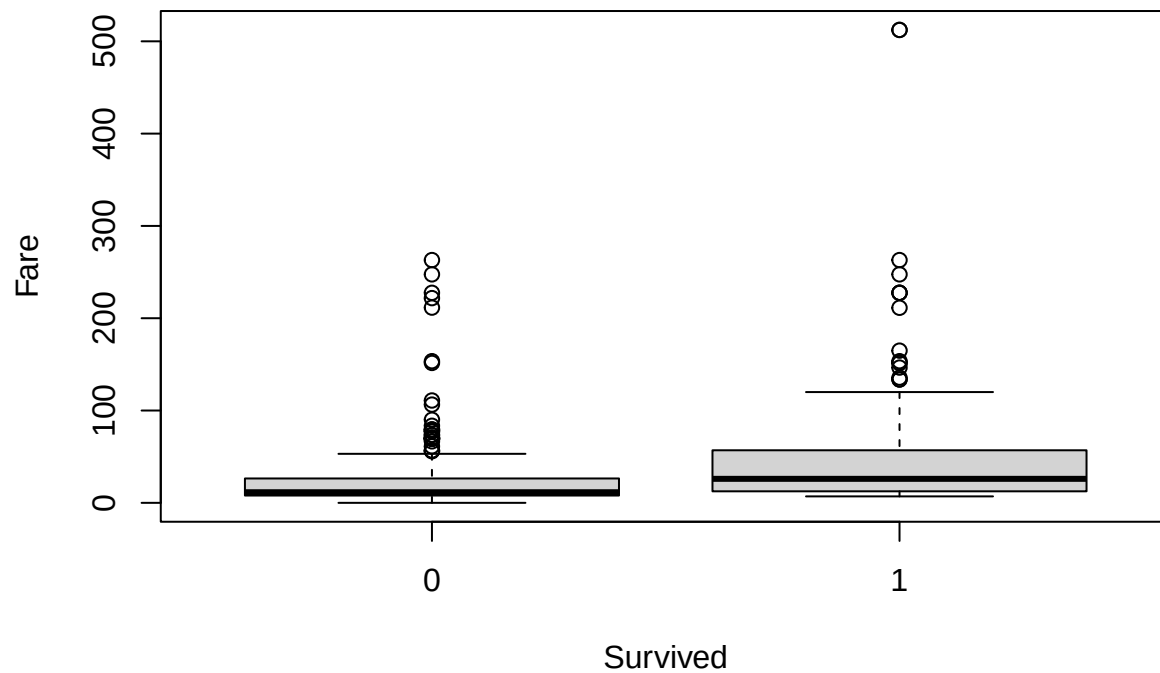
```
boxplot(Parch ~ Survived,
        data = titanic_train,
        main = 'Total Number of Parents and Children on Board')
```

Total Number of Parents and Children on Board



```
boxplot(Fare ~ Survived,  
        data = titanic_train,  
        main = 'Fare (equivalent to modern British pounds)')
```

Fare (equivalent to modern British pounds)



In the first boxplot, there appears to be no relationship between Total number of siblings and spouses on board and survival rate as both plots look the same for survived and not survived. In the second boxplot,

there appears to be a potentially weak positive relationship with total number of parents or children on board and survival rate. Lastly, in the third boxplot, there appears to be a relationship with fare cost and chances of survival. As the fare of the ticket increases, it appears that survival rates increase.

EDA ON RELATIONSHIPS BETWEEN SURVIVAL AND THE CATEGORICAL VARIABLES

To visualize the relationship between categorical predictors and survival, we will use tables and bar graphs.

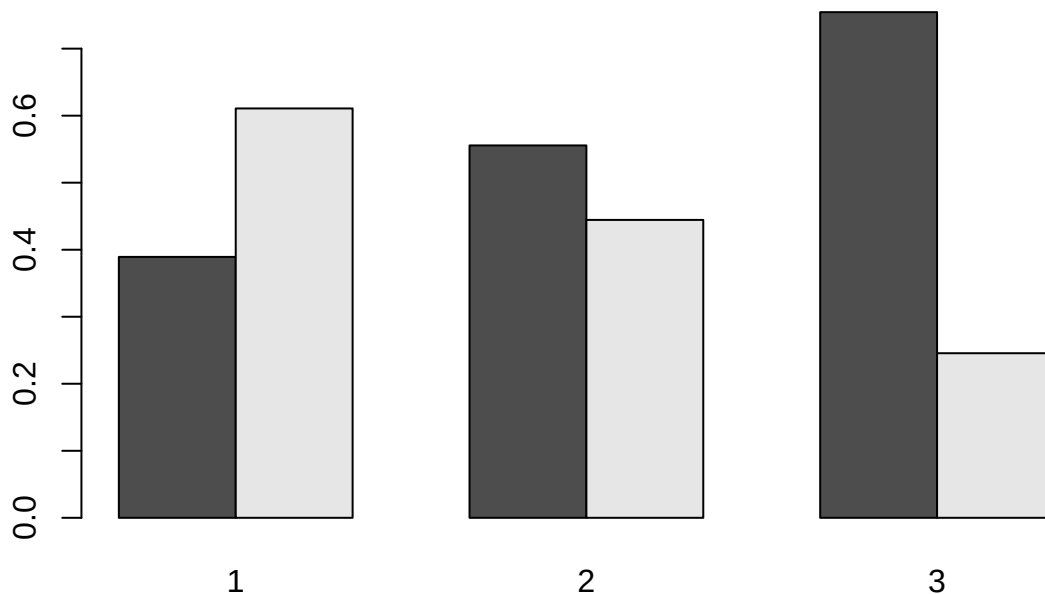
Survival by Ticket Class:

```
prop.table(table(titanic_train$Survived, titanic_train$Pclass),
            margin = 2)
```

```
##
##           1           2           3
##  0 0.3892617 0.5555556 0.7544379
##  1 0.6107383 0.4444444 0.2455621
```

```
barplot(prop.table(table(titanic_train$Survived, titanic_train$Pclass),
                        margin = 2),
        beside = TRUE,
        main = "Barplot of Survivors By Passenger Class")
```

Barplot of Survivors By Passenger Class



Survival by Port of Embarkment:

```
prop.table(table(titanic_train$Survived, titanic_train$Embarked),
            margin = 2)
```

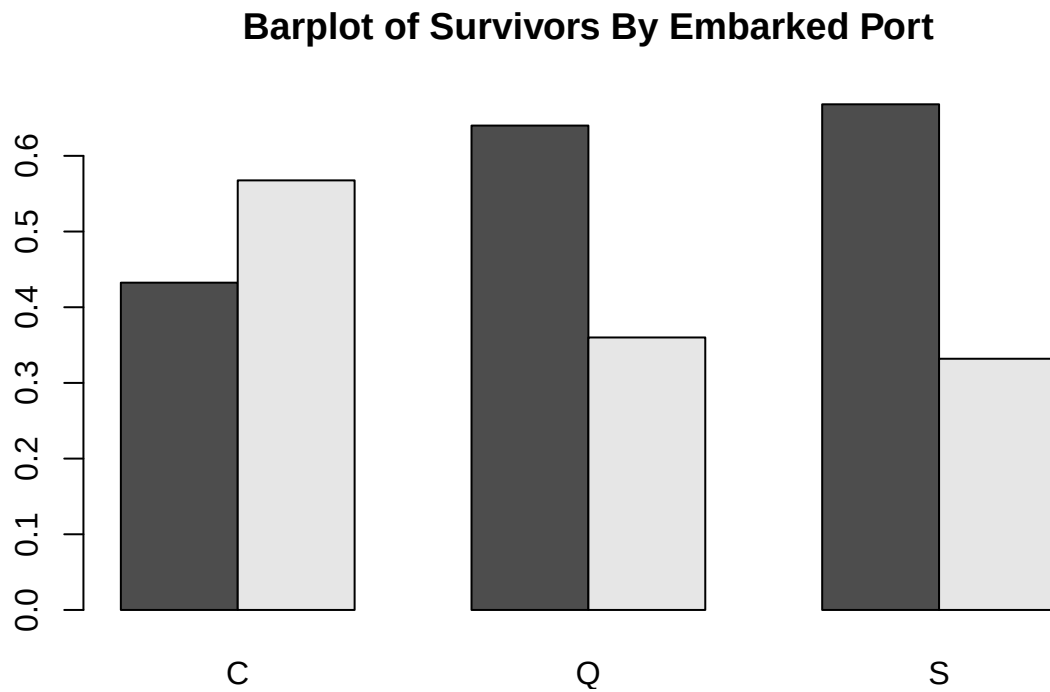
```
##
##           C           Q           S
##  0 0.4324324 0.6400000 0.6681128
##  1 0.5675676 0.3600000 0.3318872
```

```
barplot(prop.table(table(titanic_train$Survived, titanic_train$Embarked),
                        margin = 2),
```

```

beside = TRUE,
main = "Barplot of Survivors By Embarked Port")

```



Survival Rate by Gender:

```

prop.table(table(titanic_train$Survived, titanic_train$Gender),
margin = 2)

```

```

##
##      female      male
## 0 0.2512077 0.8096386
## 1 0.7487923 0.1903614

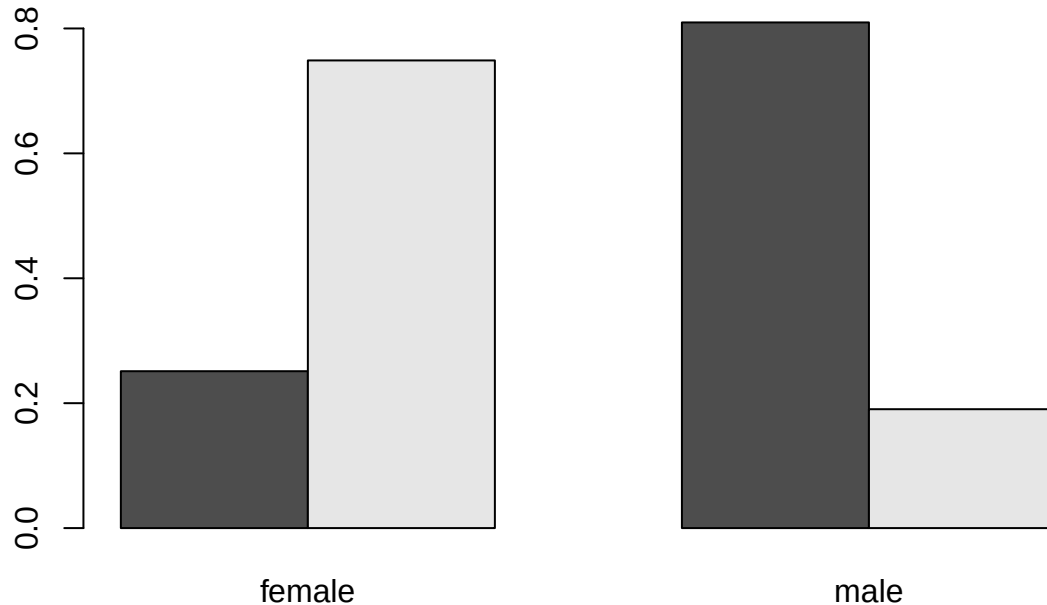
```

```

barplot(prop.table(table(titanic_train$Survived, titanic_train$Gender),
margin = 2),
beside = TRUE,
main = "Barplot of Survivors By Gender")

```

Barplot of Survivors By Gender



Looking at the bar plot and table for class, we can see that the higher the class of the passenger, the higher their chance of survival was. 61% of first class passengers survived, while only 25% of third class passengers survived.

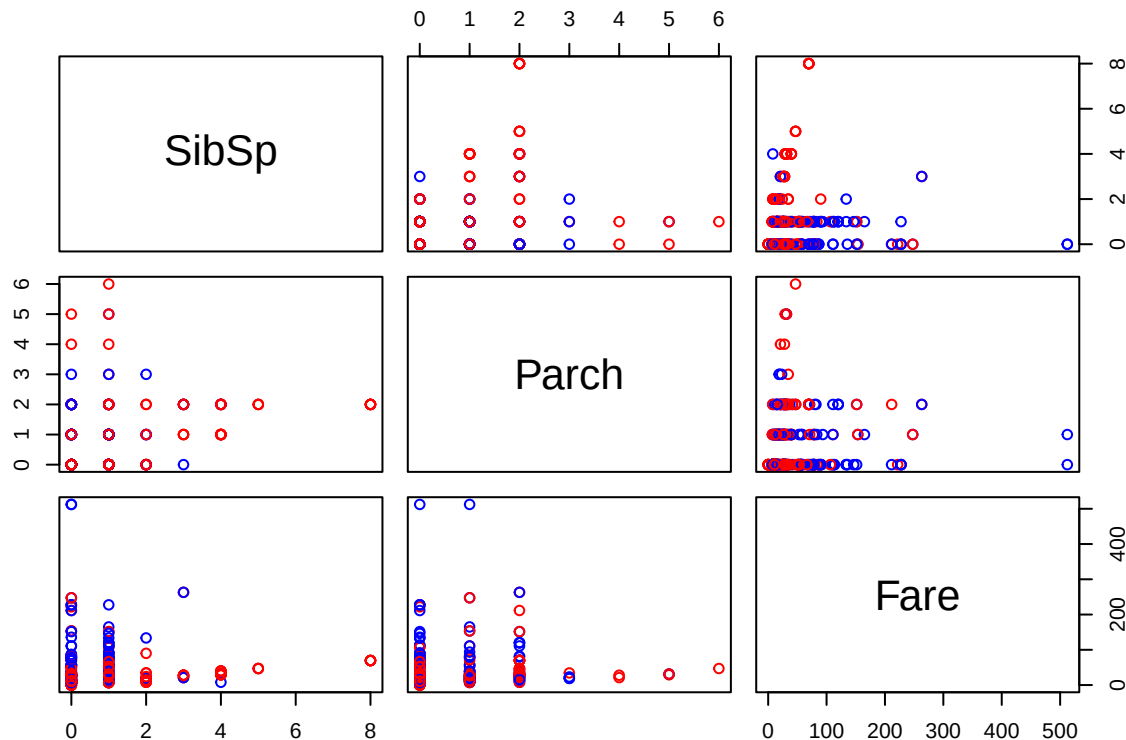
For embarkment point, looking at the bar plot and table, those who embarked from Cherbourg had the highest survival rate of around 56%, compared to Queenstown (36%) and Southampton(33%).

Looking at survival rates by gender, we can see a very large difference in survival rates; 75% of women survived, while only 20% of men survived.

SOME VISUAL EDA ON CLASSIFICATION PAIRS

Lastly, to explore which pairs of quantitative explanatory variables might help classify whether a passenger survived or not, we can construct a pairs plot.

```
pairs(titanic_train[, c(3,4,5)],  
      col = ifelse(titanic_train$Survived == 1, "blue", "red"))
```



Looking at our pairs plot, red passengers symbolize those who did not survive, while blue passengers symbolize those who did survive. We can see that Fare seems to show relevant separation which could be used to identify useful combinations, while SibSp and Parch do not.

Modeling

We will now build our models to classify and predict survival. The models we will use are linear discriminate analysis (LDA), quadratic discriminant analysis (QDA), classification trees, and binary logistic regression model.

To avoid overfitting of our models to our data, we will be using a test dataset, `titanic_test`, along with our training data, `titanic_train`.

LINEAR DISCRIMINATE ANALYSIS (LDA)

For our LDA and QDA, we will only use our quantitative variables of SibSp, Parch, and Fare. The LDA is built as followed:

```
titanic_lda <- lda(factor(Survived) ~ SibSp + Parch + Fare,
  data = titanic_train)
titanic_lda_pred <- predict(titanic_lda,
  as.data.frame(titanic_test))
table(titanic_lda_pred$class, titanic_test$Survived)
```

```
##
##      0      1
##  0 149   83
##  1  12   23
##
## (83+12)/267
## [1] 0.3558052
```

```
83/106
```

```
## [1] 0.7830189
```

```
12/161
```

```
## [1] 0.07453416
```

The LDA model returns an overall error rate of 0.356, which is a little high. We are also concerned about the very high .783 error rate when predicting survivors. However, the error for predicting non-survivors is relatively accurate with a low .074 error rate. The LDA model is good for predicting who did not survive, but should not be used to try and predict who survived.

QUADRATIC DISCRIMINANT ANALYSIS (QDA)

The QDA is built as follows:

```
titanic_qda <- qda(factor(Survived) ~ SibSp + Parch + Fare,
                  data = titanic_train)
titanic_qda_pred <- predict(titanic_qda,
                           as.data.frame(titanic_test))
table(titanic_qda_pred$class, titanic_test$Survived)
```

```
##
##      0      1
##  0 146    73
##  1   15    33
```

```
(15+73)/267
```

```
## [1] 0.329588
```

```
73/106
```

```
## [1] 0.6886792
```

```
15/161
```

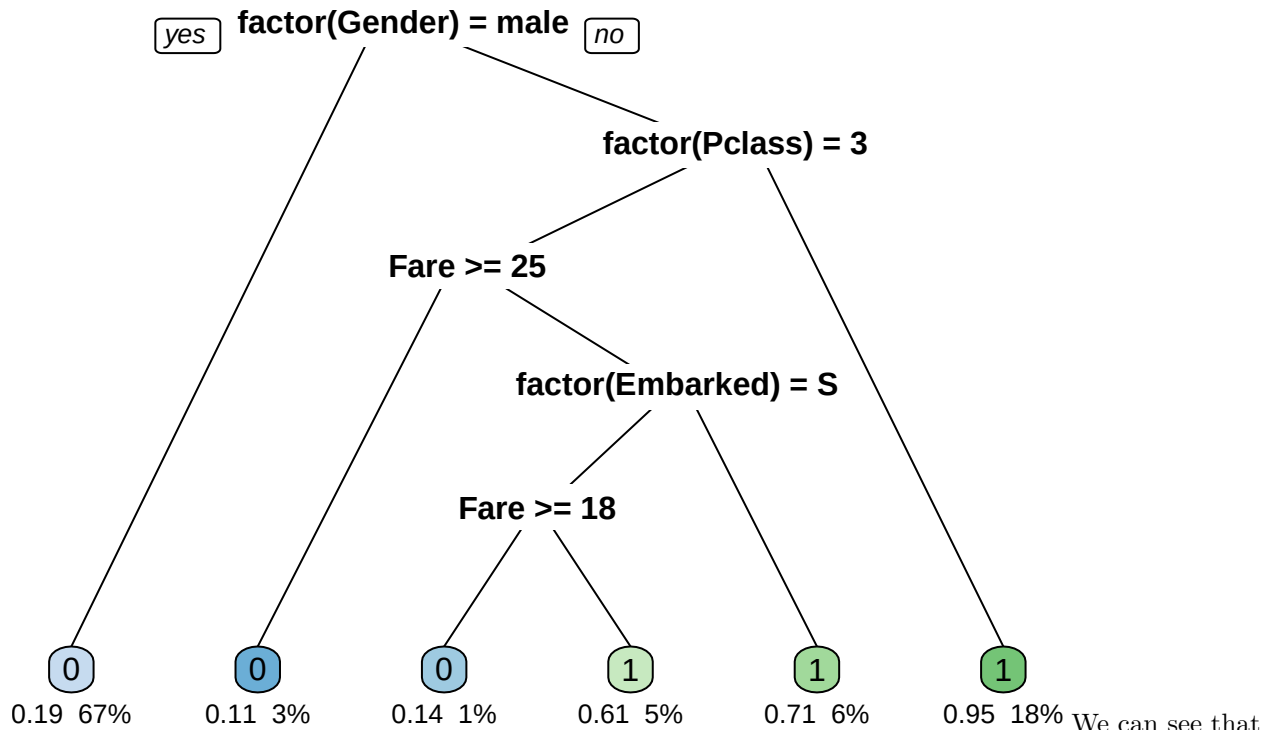
```
## [1] 0.0931677
```

As we can see, the error rate for the QDA is very similar to the LDA, but performs slightly better. The overall error rate is around a relatively high .33, which is very slightly lower than the LDA's overall error rate. The survivor predictability is very inaccurate, with a .689 error rate, while the QDA non-survivor error rate is low, at 0.093. Again, we could use the QDA model to predict an individual who did not survive, but not to predict if someone did survive.

CLASSIFICATION TREES

Utilizing a classification tree allows to take into account categorical variables into our model. Using the training data, we construct it as follows:

```
titanic_tree <- rpart(factor(Survived) ~ factor(Pclass) + SibSp + Parch + Fare
                     + factor(Embarked) + factor(Gender),
                     data = titanic_train,
                     method = 'class')
rpart.plot(titanic_tree,
           type = 0,
           clip.right.labs = FALSE,
           branch = 0.1,
           under = TRUE)
```

We can see that R determined that Gender was the most important factor in determining whether someone survived or died. The next important was Passenger Class, and then Fare, and then point of Embarkment, and then Fare again. Sibsp and Parch were not used, similar to our predictions from the EDA of classification pairs.

To check how accurate our classification tree is, we do the following:

```
titanic_tree_pred <- predict(titanic_tree,
                             as.data.frame(titanic_test),
                             type = 'class')
table(titanic_tree_pred, titanic_test$Survived)
```

```
##
## titanic_tree_pred  0   1
##                   0 141  32
##                   1  20  74
```

```
(20+32)/267
```

```
## [1] 0.1947566
```

```
32/106
```

```
## [1] 0.3018868
```

```
20/161
```

```
## [1] 0.1242236
```

The overall error rate for the Classification Trees model is significantly lower than the QDA and LDA models, at roughly .195. The survivor predictability error rate is also much lower than the QDA and LDA models, although still relatively high, at .302. The non survivor predictability error rate is slightly higher than the QDA and LDA models, at .124, but it is still relatively low on average. The classification tree model is a better model overall to use, and better to use for predicting survivors. However, the LDA model is the best for predicting non-survivors.

BINARY LOGISTIC REGRESSION

Lastly, we will consider a Binary Logistic Regression model to predict whether a passenger survived or not. Similarly to our Classification Tree model, we can use all quantitative and categorical predictors in this model.

We will train our logistic regression on training data, and then inspect the resulting confusion matrix from the test data, as follows:

```
titanic_logit <- glm(factor(Survived) ~ factor(Pclass) + SibSp + Parch + Fare
                    + factor(Embarked) + factor(Gender),
                    data = titanic_train,
                    family = binomial(link = 'logit'))
titanic_logit_prob <- predict(titanic_logit,
                             as.data.frame(titanic_test),
                             type = 'response')
```

The logistic model will yield probabilities, not a binary output (survivor/non-survivor). We will determine if the passenger survived or died by establishing a threshold probability. If $P > 0.5$, then we will say that they survived. We need to confirm that R will use the appropriate “type” for a probability greater than 0.5 being a survivor. We can run “levels” on the factored variable, as follows:

```
levels(factor(titanic_test$Survived))
```

```
## [1] "0" "1"
```

We then obtain test classification from the logistic model using a threshold probability of 0.5, as follows:

```
titanic_logit_pred <- ifelse(titanic_logit_prob > 0.5, '1', '0')
table(titanic_logit_pred, titanic_test$Survived)
```

```
##
## titanic_logit_pred  0   1
##                   0 131  30
##                   1  30  76
```

```
(30+30)/267
```

```
## [1] 0.2247191
```

```
30/106
```

```
## [1] 0.2830189
```

```
30/161
```

```
## [1] 0.1863354
```

The overall error rate of the Binary Logistic Model is about 0.225, which is better than LDA and QDA, but slightly worse than the Classification Tree. The predictability error rate for determining if someone survived is 0.283, which is the lowest of all four models, yet still relatively high. The prediction error rate for determining a non-survivor is 0.186, which is the highest error rate of all four models. This model could be used to when attempting to predict if someone survived.

FINAL RECOMMENDATIONS

Overall, the classification tree had the lowest overall error rate, at 0.194. The lowest error rate for predicting survivors is the binary logistic regression model, at 0.283. The lowest error rate for predicting non-survivors is the LDA, at an error rate of 0.074.

We recommend using the classification tree because it avoids the very high error rates in predicting survivors that the LDA and QDA models output, while having the overall lowest error rate compared to the other three classifiers.

Discussion

While the above variables used in our four classifiers do predict whether a passenger survived or died, there are other unexplored variables that should be tested in future research, such as Age, Nationality, and Physical health. The difference in gender survival rate is also very important to note. The importance of Passenger Class also highlights the sad reality that those of lower socioeconomic status are often ignored and neglected, especially in emergencies and tragedies throughout history.

Additionally, while we recommend using a classification tree model for prediction, it is important to note that none of the four models gave perfect results.